

Chapter 11

The Human Eye and the Colourful World

1. The human eye forms an image on the:

- A) Cornea
- B) Pupil
- C) Retina
- D) Iris

Answer: C) Retina

2. The outermost transparent layer of the eye is called:

- A) Iris
- B) Cornea
- C) Lens
- D) Retina

Answer: B) Cornea

3. The black circular opening in the iris is called:

- A) Retina
- B) Pupil
- C) Cornea
- D) Lens

Answer: B) Pupil

4. The coloured part of the eye that controls the size of the pupil is:

- A) Cornea
- B) Iris
- C) Retina
- D) Lens

Answer: B) Iris

5. The eye lens is made of:

- A) Plastic
- B) Water
- C) Transparent, flexible material (protein)
- D) Bone

Answer: C) Transparent, flexible material (protein)

6. The ability of the eye lens to adjust its focal length is called:

- A) Refraction
- B) Accommodation
- C) Dispersion
- D) Power

Answer: B) Accommodation

7. The image formed on the retina is:

- A) Real and erect
- B) Virtual and erect
- C) Real and inverted
- D) Virtual and inverted

Answer: C) Real and inverted

8. The least distance of distinct vision for a normal human eye is about:

- A) 10 cm
- B) 15 cm
- C) 25 cm
- D) 50 cm

Answer: C) 25 cm

9. The far point of a normal eye is:

- A) 25 cm
- B) Infinity
- C) 1 m
- D) 50 m

Answer: B) Infinity

10. The defect in which a person cannot see distant objects clearly is called:

- A) Hypermetropia
- B) Myopia
- C) Presbyopia
- D) Astigmatism

Answer: B) Myopia

11. Myopia can be corrected by using:

- A) Convex lens
- B) Concave lens
- C) Cylindrical lens

D) None of these

Answer: B) Concave lens

12. The defect in which a person cannot see nearby objects clearly is called:

A) Myopia

B) Hypermetropia

C) Presbyopia

D) Astigmatism

Answer: B) Hypermetropia

13. Hypermetropia can be corrected by using:

A) Convex lens

B) Concave lens

C) Cylindrical lens

D) None of these

Answer: A) Convex lens

14. The defect of vision caused due to loss of accommodation with ageing is:

A) Myopia

B) Hypermetropia

C) Presbyopia

D) Astigmatism

Answer: C) Presbyopia

15. Presbyopia can be corrected by using:

A) Bifocal lens

B) Convex lens

C) Concave lens

D) Cylindrical lens

Answer: A) Bifocal lens

16. The ability of the eye to focus on near and far objects is called:

A) Power of accommodation

B) Power of lens

C) Refraction

D) Persistence of vision

Answer: A) Power of accommodation

17. The splitting of white light into its component colours is called:

- A) Reflection**
- B) Refraction**
- C) Dispersion**
- D) Scattering**

Answer: C) Dispersion

18. The band of seven colours formed after dispersion is called:

- A) Spectrum**
- B) Rainbow**
- C) Halo**
- D) Mirage**

Answer: A) Spectrum

19. The dispersion of light was first studied by:

- A) Newton**
- B) Einstein**
- C) Galileo**
- D) Rutherford**

Answer: A) Newton

20. The phenomenon responsible for the formation of a rainbow is:

- A) Reflection**
- B) Refraction and dispersion**
- C) Diffraction**
- D) Scattering**

Answer: B) Refraction and dispersion

21. The splitting of sunlight by water droplets in the atmosphere forms a:

- A) Halo**
- B) Rainbow**
- C) Mirage**
- D) Spectrum**

Answer: B) Rainbow

22. The red colour of the sun at sunrise and sunset is due to:

- A) Dispersion of light**
- B) Scattering of light**
- C) Reflection of light**

D) Refraction of light

Answer: B) Scattering of light

23. The blue colour of the sky is due to:

A) Reflection

B) Refraction

C) Scattering of light

D) Dispersion

Answer: C) Scattering of light

24. Which colour is scattered the most?

A) Red

B) Green

C) Blue

D) Violet

Answer: D) Violet

25. Which colour is scattered the least?

A) Red

B) Blue

C) Green

D) Violet

Answer: A) Red

26. The phenomenon responsible for twinkling of stars is:

A) Reflection

B) Refraction of light through atmosphere

C) Dispersion

D) Diffraction

Answer: B) Refraction of light through atmosphere

27. The Tyndall effect is due to:

A) Absorption of light

B) Scattering of light by colloidal particles

C) Reflection of light

D) Refraction of light

Answer: B) Scattering of light by colloidal particles

28. The sky appears dark to astronauts because:

- A) There is no sunlight in space
- B) There is no atmosphere to scatter light
- C) Eyes cannot see in space
- D) Sunlight is absorbed

Answer: B) There is no atmosphere to scatter light

29. The phenomenon of advance sunrise and delayed sunset is due to:

- A) Scattering
- B) Refraction of sunlight through atmosphere
- C) Dispersion
- D) Reflection

Answer: B) Refraction of sunlight through atmosphere

30. The persistence of vision in human eye is about:

- A) 1/10 second
- B) 1/16 second
- C) 1/25 second
- D) 1 second

Answer: B) 1/16 second
